

Stafny Velitia Karkada

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TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C, C++

Frontend: React, Redux(Saga/Thunk), React Native, Vue, Next.js, HTML5, CSS, SASS, ES6, CMS(Contentful,Drupal), Bootstrap, Jest, Tailwind, Enzyme, React Testing Library, Cypress, Playwright, axe-core, WAVE, MUI, Figma, Deque, WCAG 2.2

Backend & Infra: Node.js, Express, REST APIs, GraphQL, PostgreSQL, MongoDB, Firebase, Oracle DB, SQL, Neo4j, FastAPI, Docker, Kubernetes, AWS, GCP, Jenkins, GitHub Actions, OAuth 2.0, JWT, Google Maps Platform, Jira, Confluence

AI & ML: RAG, PineCone, TensorFlow, PyTorch, RNNs, Transformers, LangChain, SSE, Amazon Bedrock, Lovable, Bolt

EXPERIENCE

Software Engineer Frontend, Dune Security (NYC, USA)

Dec 2025 - Present

- Built and scaled Dune Security's core product from scratch using Next.js and TypeScript, restructuring the codebase and migrating components to a typed architecture to support 2x team and product growth.
- Improved application performance and reliability by introducing a client-side caching layer with TanStack Query, optimizing data-fetching and state management; Built and productionized robust engineering workflows by adding comprehensive unit tests and end-to-end automation using Selenium.

Software Engineer, Saayam For All (SF, USA)

Jun 2025 - Nov 2025

- Designed a full-stack RESTful application with JWT-based authorization, PayPal OAuth integration, and PostgreSQL schema with re-executable DDL/DML scripts and PL/pgSQL stored procedures/triggers for consistent deployments.
- Implemented Twelve-Factor principles on AWS, provisioning cloud infrastructure via Pulumi. Automated CI/CD with GitHub Actions, integrated CloudWatch metrics, and auto-scaling with AWS Load Balancer.
- Built C++ utilities leveraging pointer-based data structures for backend performance and memory analysis.

UX Engineer, Autodesk (SF, USA)

Apr 2024 - Dec 2024

- Led Accessibility audit of the Autodesk Developer Portal, identified 100+ issues using Axe DevTools, WAVE, and Accessibility Insights, and fixed violations with axe-core, axe-jest, Playwright, and Cypress.
- Designed and implemented a new team-management workflow using React,Drupal,Material UI, enabling role-based access, consistent state handling,increasing developer engagement by 60% (Pendo).
- Identified recurring accessibility issues across the portal, traced them to the design system, and contributed upstream fixes; authored centralised documentation and led workshops to drive adoption of ally standards across teams.

Software Engineer,Y Media Labs (Bangalore, India)

Aug 2020 - Aug 2023

- Revamped California Pizza Kitchen's web platform using React Hooks(TypeScript), lazy loading, cutting user clicks by 75%, boosting conversions 2.5x, doubling loyalty memberships. Enhanced SEO and Core Web Vitals using Lighthouse audits and CI-integrated metrics reporting, recognized by their CMO.
- Part of the core team building YML's first design system from scratch, collaborating with leadership and clients. Conducted research, prototyped with Storybook, and developed reusable React components and design tokens, standardizing UI patterns, increasing production efficiency by 70% and reducing team size requirements by 60%.
- Built a full-stack white-labeled B2B web solution for Clover Payments using React, Node.js, and MongoDB, enabling rebranding for 20+ clients. Wrote unit tests with Jest and React Testing Library to ensure reliability.
- Built a CMS-integrated Vue.js site for WeTransfer using GraphQL and Contentful, enabling the marketing team to manage content independently using headless CMS, reducing developer dependency and accelerating time-to-market.
- Built an in-house React Native mobile app for health tracking, using Firebase Firestore, Firebase Cloud Messaging (FCM) for push notifications, and Google Analytics to track feature adoption and engagement metrics.
- Deployed a data-heavy website using Redux Sagas for state management, implemented efficient pagination and client-side caching to optimize load times. Set up SonarQube in the Jenkins CI/CD pipeline for code quality checks.
- Led team of 4 to integrate GPT-3 into company website, coordinating with design stakeholders and presenting technical roadmaps to clients, enabling AI-driven dynamic user responses with real-time API integration.

Graduate Web Teaching Assistant, Northeastern University (Boston, MA)

Jan 2024 - Aug 2025

- Delivered live coding workshops and provided real-time debugging support while mentoring 140+ students and 50+ project teams in building end-to-end, multi-user applications with WebSockets, authentication, and AI integrations.

PROJECTS & ACADEMIC RESEARCH

HealthPlan API Platform | Elasticsearch, Redis, RabbitMQ, Kibana, JWT, Node.js, Express

- Designed and implemented a scalable health data platform using Node.js, Redis, and Elasticsearch, exposing RESTful APIs with JSON Schema validation, Dockerized services, and an asynchronous RabbitMQ messaging pipeline.

Wyckoff AI Trading Bot | Next.js ,TypeScript, Tailwind CSS, Python, LLMs

- Developed a conversational GenAI application using OpenAI's models and vector embeddings, implementing RAG architecture; built with Next.js and FastAPI to support server-sent events (SSE) for real-time chat streaming.
- Experimented with Neo4j Bloom and Amazon Bedrock to enhance chatbot capabilities, leveraging Neo4j Knowledge Graph to model trading concepts and deliver graph-based insights.

EDUCATION

MS in Computer Science: Northeastern University (Boston, MA,USA)

Apr 2025

Bachelor of Engineering in Computer Science: NMAM Institute Of Technology (Karnataka, India)

Aug 2020